

Refactor of the cohomology passage, v3: standard language, free-vs-purchased relations

Editing companion; supersedes v1/v2

June 17, 2026

How to use this file. Block 1 replaces lines 313–351 (delete the answered %DAM note at 321; line 353 follows directly). Block 2 replaces lines 767–774 in Section `sec:cocycle` (delete the %DAM note at 774). Block 3 is a one-line sign fix at `eq:bp-cocycle-coboundary`. Copy what lies between the % ==== BEGIN/END PASTE ==== markers in the source. `\ref/\cite` target the main file and render here as “??”/“[?]” by design. The source header lists the lines in the main file where leftover “gauge” language should be swept for consistency (notably the conclusions block at 3060–3079, which mirrors the old intro).

Block 1 — replaces lines 313–351 (intro, §1.1)

The price of trading one divergent contour for a family of finite ones is redundancy: each $C_\gamma^f(\tau_0)$ depends on an auxiliary basepoint τ_0 and on a group element γ , while the period polynomial must refer to neither. Degree-one group cohomology is the standard bookkeeping for exactly this situation, and only three definitions are needed. A 1-*cocycle* is a function $c: \Gamma \rightarrow V_n$ satisfying $c_{gh} = c_g|_h + c_h$, and $\gamma \mapsto C_\gamma^f(\tau_0)$ is one for each fixed τ_0 (Lemma ??). A 1-*coboundary* is the special cocycle $\delta P: \gamma \mapsto P|_\gamma - P$ built from a polynomial $P \in V_n$; by (??), shifting $\tau_0 \rightarrow \tau'_0$ (no pole crossed) shifts the whole family $\{C_\gamma^f\}_\gamma$ by one such δP — the same P for every γ , which is why the quotient retains anything at all. The *cohomology class* $[C^f] \in H^1(\Gamma; V_n) := \{\text{cocycles}\}/\{\text{coboundaries}\}$ is therefore basepoint-free, and a *representative* of it is any cocycle $C^f - \delta P$. This kills the τ_0 .

A reader outside the field might object that no machinery should be needed: the classical period polynomial is a single integral along one S -contour, so why not take the finite S -contour $C_S^f(\tau_0)$ and stop? Because at a finite basepoint the raw S -value is pure redundancy. Torsion alone forces $C_S^f|(1+S) = C_{S^2}^f = C_{-I}^f = 0$ at *every* τ_0 , and since S acts on V_n as an involution, its (-1) -eigenspace is also the image of δ_S : the value C_S^f lies in $\ker(1+S) = \text{im } \delta_S$, exactly the space of shifts that basepoint motion induces on it. (At $k = 12$ this kernel is six-dimensional inside the eleven-dimensional V_{10} ; membership selects a subspace, never a point.) The two relations cutting out the Kohnen–Zagier space W_{k-2} [?] therefore have different statuses: the S -relation is *free*, holding before any choice is made, while the U -relation must be *purchased* — it appears only after the representative is pinned, and pinning is the role of the other generator.

The generator T is built for the job. It has infinite order and acts on V_n unipotently — a single Jordan block — so $\delta_T := (\cdot)|_T - \text{id}$ has one-dimensional kernel and cokernel: the equation has one scalar of obstruction and one scalar of slack. We choose, within the class, the representative $C^f - \delta P$ whose T -value vanishes, i.e. solve $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$ on V_n . Solvability is itself arithmetic: unit-segment T -periodicity identifies the obstruction in $V_n / \text{im } \delta_T \cong \mathbb{C} \cdot [X^n]$ as the constant Fourier

mode (Lemma ??), $[C_T^f]_{X^n} = (2\pi i)^{n+1} c_f(0)$ — the residue of ω_f at the very cusp that T stabilizes. The normalized representative exists if and only if $c_f(0) = 0$; classes that pass are the *parabolic* ones, identified by Eichler–Shimura [?] with $S_k \oplus \overline{S_k}$, the lone obstructed direction being Eisenstein. Cuspidality, in this framework, is not a convergence hypothesis; it is the solvability condition for the normalization.

With the T -value normalized away, the entire cuspidal content sits on the surviving generator value,

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P), \quad (1)$$

well-defined up to the line $\mathbb{C}(X^n - Y^n)$ left over from $\ker \delta_T = \mathbb{C}Y^n$ — an ambiguity touching only the boundary monomials, which is why Theorem ?? ranges over the interior ℓ alone. The U -relation now arrives, purchased as promised: conditioned on the vanishing T -value, the group relation $(ST)^3 = -I$ becomes $r_f | (1 + (TS) + (TS)^2) = 0$, and r_f is the period polynomial. Indeed, for holomorphic $f \in S_k$, sending $\tau_0 \rightarrow i\infty$ kills C_T^f outright and (1) collapses to the classical integral (??) — whose familiar freedom from τ_0 and γ is thereby revealed as the silent choices $\gamma = S$, $\tau_0 = i\infty$. Its interior coefficients are the periods against Brown’s rational basis, $[r_f]_{X^{k-2-\ell}Y^\ell} = \omega^\pm(f) c_S^\ell$; the precise coordinate statement is given in Section ??. The finite S -contour was the right object all along, but it requires the correction term in (1), sourced entirely by the T -contour — and producing that correction in closed form is the business of this paper.

The existing line 353, “The coboundary equation admits a closed-form inverse — the polynomial inversion of [DR2017]. . .”, follows the final sentence above directly; “that correction in closed form” is its antecedent. For meromorphic f with interior poles, the identity $C_S^f(1 + S) = 0$ holds per homotopy component; state that precisely in Section sec:wall-crossing, not here.

Block 2 — replaces lines 767–774 (Section sec:cocycle)

Parabolic cohomology and periods. The cohomology class $[C^f] \in H^1(\Gamma; V_n)$ is independent of τ_0 (on the connected component) and of the choice of δP representative. Its restriction to the cusp stabilizer $\langle T \rangle \cong \mathbb{Z}$ lands in $H^1(\langle T \rangle; V_n) = V_n / \text{im } \delta_T \cong \mathbb{C} \cdot [X^n]$, and by Lemma ??(c) the restricted class is $(2\pi i)^{n+1} c_f(0) \cdot [X^n]$: the residue of ω_f at the cusp, valued in the T -coinvariants. The class is *parabolic* — cohomologous to a cocycle with vanishing T -value — if and only if this residue vanishes, i.e. if and only if $f \in S_k^!$. The classical Eichler–Shimura theorem [?] identifies $H^1(\Gamma; V_n) \cong S_k \oplus \overline{S_k} \oplus \mathbb{C}_{\text{Eis}}$ and the parabolic subspace $H_{\text{par}}^1(\Gamma; V_n) \cong S_k \oplus \overline{S_k}$; on cocycles normalized so that $c_T = 0$, the assignment $c \mapsto c_S$ further identifies $H_{\text{par}}^1(\Gamma; V_n) \cong W_{k-2} / \mathbb{C}(X^n - Y^n)$, with the quotient line accounting for the residual freedom $P \mapsto P + \mathbb{C}Y^n$. When $\dim S_k = 1$, each parity eigenspace of H_{par}^1 is one-dimensional, and the two coordinates of a parabolic class in Brown’s rational basis $p_f^\pm$ are its periods: $[C^f]^\pm = \omega^\pm(f) [p_f^\pm]$ for $f \in S_k$, while the quasi-periods $\eta^\pm$ arise as the coordinates of the class $[C^{f'}]$ of a weakly holomorphic Hecke partner $f' \in S_k^!$ (at $k = 12$: $f = \Delta$, $f' = \hat{\Delta}$) [?].

Block 3 — sign fix in eq:bp-cocycle-coboundary (lines 755–760)

Minimal fix: in the final clause of the display, swap the arguments of F , replacing $P = (2\pi i)^{n+1}(F(\tau'_0) - F(\tau_0))$ by

$$P = (2\pi i)^{n+1}(F(\tau_0) - F(\tau'_0)) \in V_n \otimes \mathbb{C}.$$

Derivation (two lines, via the equivariance step of Lemma ??, $\int_{\gamma^{-1}a}^{\gamma^{-1}b} \omega_f = (\int_a^b \omega_f)|_\gamma$): splitting $\int_{\gamma^{-1}\tau'_0}^{\tau'_0} = \int_{\gamma^{-1}\tau'_0}^{\gamma^{-1}\tau_0} + \int_{\gamma^{-1}\tau_0}^{\tau_0} + \int_{\tau_0}^{\tau'_0}$ gives

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (2\pi i)^{n+1} \left[\left(\int_{\tau'_0}^{\tau_0} \omega_f \right) \Big|_\gamma + \int_{\tau_0}^{\tau'_0} \omega_f \right] = A - A|_\gamma, \quad A := (2\pi i)^{n+1} \int_{\tau_0}^{\tau'_0} \omega_f,$$

which is $P|_\gamma - P$ for $P = -A$. The cohomological conclusion is unaffected; only the labelling of P changes. Check the convention once at the single downstream point where P appears explicitly, the normalized value (1).